02/06/2026

Usability testing report – high fidelity prototypes

Human-Computer Interaction Project

Team 4

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1 Goal of evaluation

Evaluate the performance (effectiveness and efficiency) of participants when using the high-fidelity prototype. Evaluate the participants' satisfaction with the SUS questionnaire, and their user experience with the UEQ questionnaire.

2 Schedule update

Table 1. Schedule updates.

Change	Section	Explanation
1	**Dates and times**	Sessions 2 through 6 were rescheduled relative to the original plan. Session 2 moved from 12/05/2026 at 19:30 to 13/05/2026 at 12:30. Sessions 3 and 4, originally planned for 13/05/2026, were postponed to 31/05/2026 at 17:30 and 17:45 respectively. Sessions 5 and 6, originally planned for 18/05/2026, took place on 01/06/2026 at 12:30 and 14:00 respectively. These reschedulings were necessary to accommodate the availability of participants, all of whom are elderly adults with limited scheduling flexibility. No changes were made to the tasks, materials, or prototype itself.
2	**Optimal values**	The optimal values for elemental actions and completion time specified in the planning document (Report 6, 2.8.2) contained miscalculations. The correct optimum actions are 14 (Task 1) and 22 (Task 2), replacing the planned values of 27 and 18; the correct optimum times are 205 s (Task 1) and 275 s (Task 2), replacing 105 s and 80 s, and represent the expected duration for an average target-population user completing each task without errors. An additional one-time onboarding cost of 4 actions precedes both tasks and is excluded from all per-task calculations.

3 Information about the performed usability testing

3.1 Dates and places

Table 2. Dates, places, and team composition for each session.

Session	Date and time	Place	Participants and persona	Team members and roles
1	12/05/2026 20:40	Remote – Bari	1 participant (P1) Rosa – Self-Blaming Novice	Facilitator: Fabio Vito Papapicco Observers: Riccardo Cirillo, Andrea Ramaglia Bianchino
2	13/05/2026 12:30	Remote – Bari	1 participant (P2) Antonio – Proud Delegator	Facilitator: Fabio Vito Papapicco Observer: Ilaria Salami

3	31/05/2026 17:30	Remote – Bari	1 participant (P3) Rosa – Self-Blaming Novice	Facilitator: Fabio Vito Papapicco Observer: Riccardo Cirillo
4	31/05/2026 17:45	Remote – Bari	1 participant (P4) Rosa – Self-Blaming Novice	Facilitator: Fabio Vito Papapicco Observers: Ilaria Salami, Andrea Ramaglia Bianchino
5	01/06/2026 12:30	Remote – Turin	1 participant (P5) Rosa – Self-Blaming Novice	Facilitator: Riccardo Cirillo Observers: Andrea Ramaglia Bianchino
6	01/06/2026 14:00	Remote – Turin	1 participant (P6) Rosa – Self-Blaming Novice	Facilitator: Riccardo Cirillo Observers: Ilaria Salami, Andrea Ramaglia Bianchino

3.2 Participant demographics

The usability testing involved 6 participants (P1–P6), recruited through convenience sampling among family members residing in Italy. All participants represent the target user population: elderly adults who need to interact with Italian digital public services (specifically CUP medical appointment booking and pagoPA bill payment) for personal administrative tasks.

Age and gender. Ages range from 53 to 82 years (M = 66.7, SD = 10.3). P2 (82 years old) is the participant who most closely matches the core target of 70–85+, and most closely embodies the Antonio persona. P1 (74 years old) also falls within the upper portion of the target range and, in terms of delegation behaviour and digital confidence, likewise maps to the Rosa profile. The remaining four participants (53–67 years) exhibit behavioural patterns consistent with the Rosa persona. Gender is evenly balanced: 3 female (P1, P3, P5) and 3 male (P2, P4, P6). Two of the six participants had previously taken part in the low-fidelity testing phase; the partial reuse was necessary given the difficulty of remotely recruiting a sufficient number of users within the 70–85+ target range.

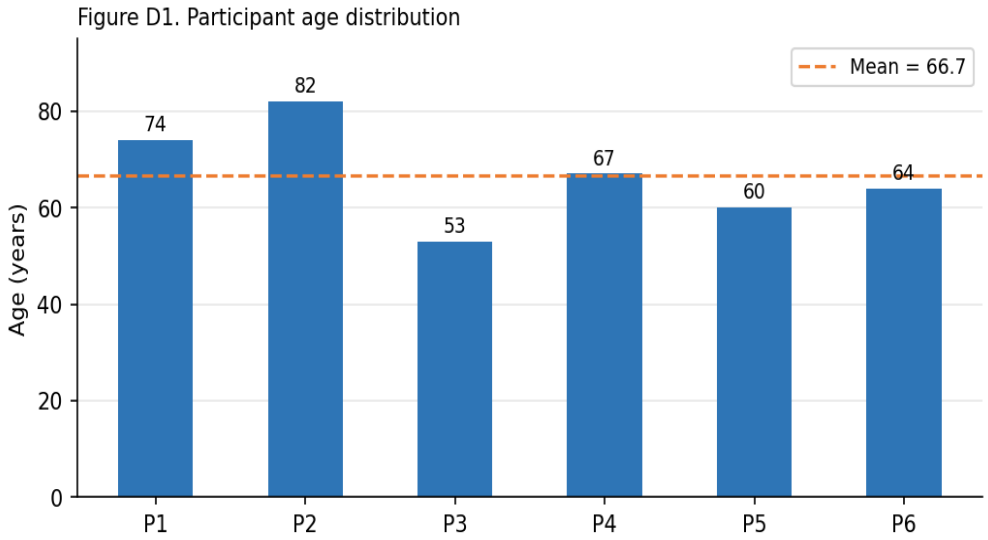


Figure 1. Participant age distribution. The dashed line indicates the group mean (M = 66.7 years).

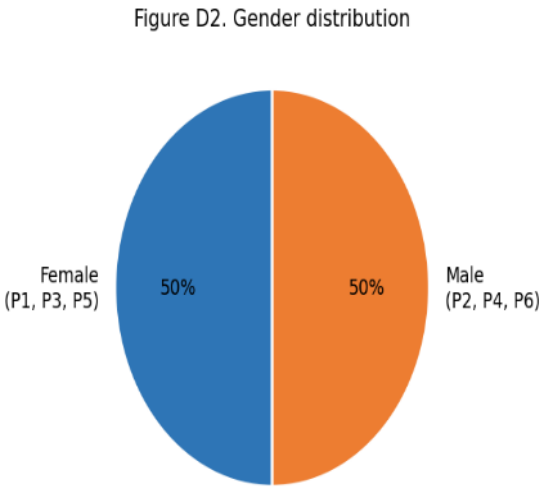


Figure 2. Gender distribution (n = 6): 50% female, 50% male.

Digital device usage. PC usage varies across participants: three (P3, P4, P5) use a computer for more than 3 hours daily, two (P2, P6) between 1 and 3 hours, and P1 never uses one. Smartphone use shows a different distribution: three participants (P1, P3, P6) use their smartphone between 1 and 3 hours per day, two (P2, P4) for less than 1 hour, and only P5 for more than 3 hours. This contrast is significant given that the Interactive Mentor prototype is designed for smartphone interaction, and confirms the "Mnemonic Silo" pattern identified during the context-of-use analysis: desktop computing competence does not readily transfer to mobile device proficiency**.**

Perceived digital confidence. Self-reported confidence scores range from 2 to 5, with a group mean of 3.17 (SD = 0.98). Four participants scored 3, P1 scored 2 (below neutral), and P5 scored 5 (fully confident). This distribution provides clear low- and high-confidence reference points useful for interpreting performance differences across the dataset.

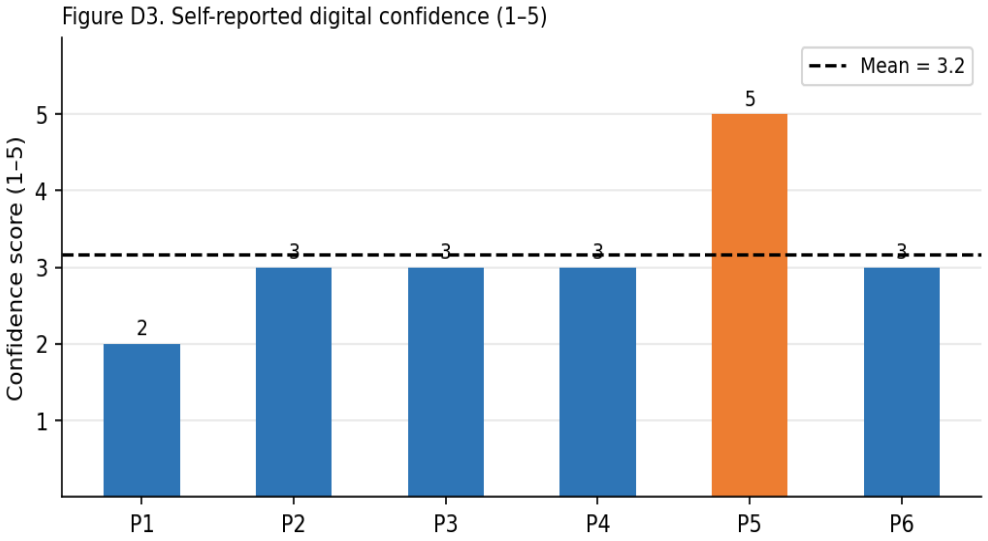


Figure 3. Self-reported digital confidence scores (scale 1–5). The dashed line indicates the group mean (M = 3.17).

Paper notebook usage. 5 out of 6 participants (83%) report using a paper notebook to record passwords or step-by-step procedures. Only P5 does not. This replicates the result observed

during the low-fidelity testing phase (5/6, 83%) and confirms the "Hybrid Workaround" coping behaviour identified in the context-of-use analysis as a stable characteristic of the target population.

Attitude toward new interfaces. 3 out of 6 participants (50%) report needing to ask for help as their primary reaction (P1, P4, P6). P5 reports curiosity (17%), P3 fear of breaking something (17%), and P2 a mixed reaction of curiosity and needing help (17%). The dominant response of needing external assistance underscores the importance of the embedded step-by-step assistant in the Interactive Mentor design.

Autonomy in CUP/pagoPA tasks. 2 out of 6 participants (33%) — P4 and P5 — report always managing these tasks independently. Three (P2, P3, P6) manage them only sometimes (50%), and P1 fully delegates to others (17%). This distribution closely mirrors the low-fidelity phase sample and confirms the recruited group is representative of the target population's real-world relationship with Italian public digital services.

4 Effectiveness

4.1 Effectiveness results

Table 3. Effectiveness results by task (n = 6).

	Mistakes (average)	**Mistakes (std. dev.)**	**Success rate**
Task 1	1.50	2.07	100% (6/6)
Task 2	1.83	1.94	100% (6/6)

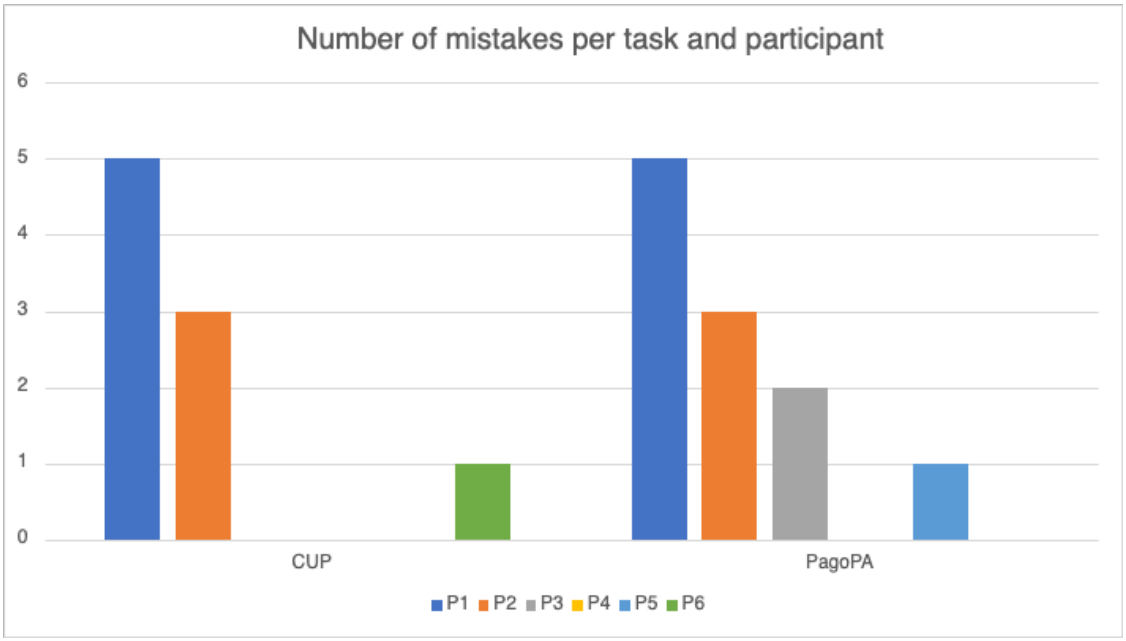


Figure 4. Number of mistakes per participant and task (n = 6).

Effectiveness findings:

- **Finding 1:** P1 and P2 together account for 16 out of 20 total mistakes across both tasks (80%); both are the oldest participants (74 and 82 years old) and reported the lowest digital confidence scores, confirming a strong correlation between age, low digital confidence, and error rate.
- **Finding 2:** Form field completion in Task 1, particularly the Codice Fiscale, is the primary error source: P1 committed 5 mistakes due to uncertainty on the required data

format, and P2 generated 3 mistakes by attempting to advance through incomplete fields.

- **Finding 3:** In Task 2, 2 out of 6 participants (33%) — P2 and P3 — selected the wrong payment method; P1 encountered an email mismatch error that could not be self-resolved and required facilitator intervention, the only instance of blocked task progress in the entire dataset.
- **Finding 4:** 3 out of 6 participants (P3, P4, P5) completed Task 1 with zero mistakes, and no task abandonment or unrecoverable error was recorded across any session; all 6 participants completed both tasks successfully.

4.2 Effectiveness analysis

Both the LoFi Mentor prototype and the HiFi Mentor prototype achieved a 100% task success rate across all sessions, making success rate a non-discriminating metric. The comparison is therefore based exclusively on the number of mistakes per task.

The high-fidelity prototype has better effectiveness than the low-fidelity prototype, because the HiFi Mentor produced fewer average mistakes in both Task 1 and Task 2. For Task 1 (CUP Booking), the HiFi average dropped to 1.50 mistakes (SD = 2.07) from the LoFi average of 1.67 (SD = 1.63), a reduction of 10%. For Task 2 (pagoPA Payment), the HiFi average dropped to 1.83 (SD = 1.94) from the LoFi average of 2.17 (SD = 2.32), a reduction of 16%. The prototype iteration therefore delivered a measurable improvement in effectiveness on both tasks.

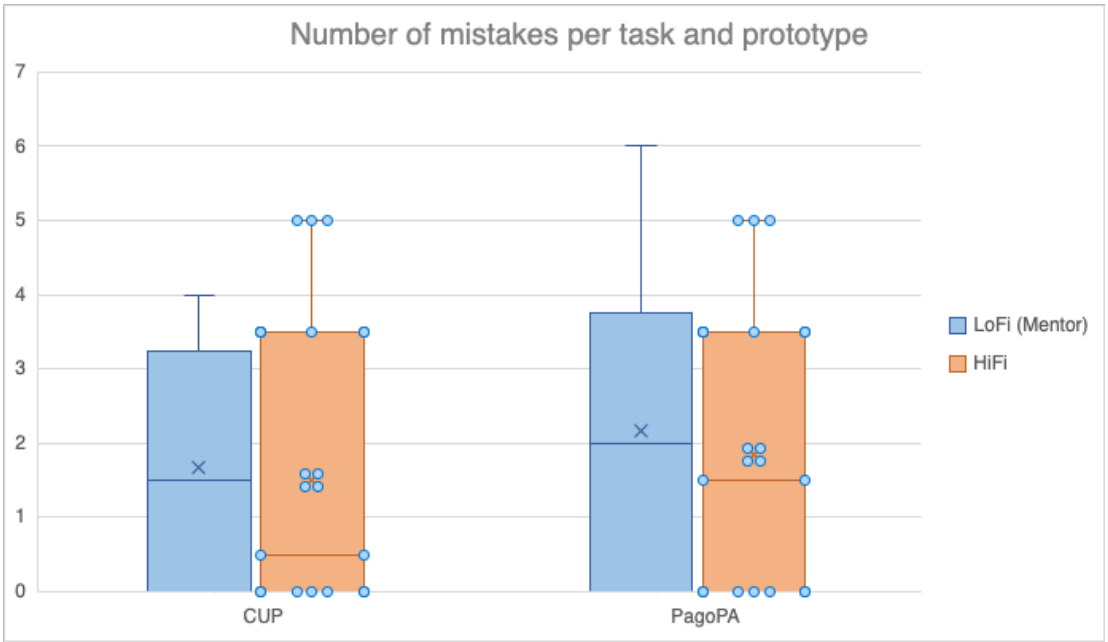


Figure 4b. Box-and-whiskers comparison of the number of mistakes per task: LoFi Mentor vs HiFi Mentor.

The box-and-whiskers comparison makes the distributional improvement visible. For Task 1, both the median and the interquartile range are lower in the HiFi than in the LoFi, though the HiFi shows a higher maximum (P1: 5 mistakes) compared to the LoFi maximum. This reflects a more polarised distribution in the HiFi: four participants made zero or one mistake while two made five each, a pattern consistent with a design that supports digitally confident users well but still presents a significant learning barrier for the least experienced ones. For Task 2, both the median and the IQR are lower in the HiFi. The reduction is attributable in part to the redesign of the payment method icons undertaken after the LoFi evaluation, which addressed one of the most consistent error sources in the LoFi Mentor phase.

Effectiveness problems of the high-fidelity prototype:

- **Problem 1:** CUP form field guidance: the Codice Fiscale field and other mandatory fields do not display format hints or completion requirements before the user attempts to advance. This caused 8 mistakes across P1 and P2 in Task 1.
- **Problem 2:** Email validation in pagoPA: when a participant (P1) entered a mismatched email address, the error was not self-resolvable and required direct facilitator intervention, blocking task progress. No self-correction mechanism or clear error guidance was triggered.
- **Problem 3:** Payment method selection in pagoPA: 2 out of 6 participants (P2, P3) selected the wrong payment method. The affordance differentiating payment options is not sufficiently prominent for first-time users.

5 Efficiency

5.1 Efficiency results

Table 4. Efficiency results – time (n = 6). Optimum time values represent the minimum expected duration for error-free task completion.

	Time (avg.)(s)	Time (std. dev.) (s)	Optimum time(s)	Time ratio
Task 1	253.2	175.1	205	1.23
Task 2	287.0	185.7	275	1.04

Table 5. Efficiency results – elemental actions (n = 6). Optimum values recalculated from the prototype's minimal interaction path, correcting a miscalculation in Report 6 (2.8.2), which reported 27 and 18 respectively.

	Actions (avg.)	Actions (std. dev.)	Optimum number of actions	Actions ratio
Task 1	17.50	8.31	14	1.25
Task 2	23.67	3.39	22	1.08

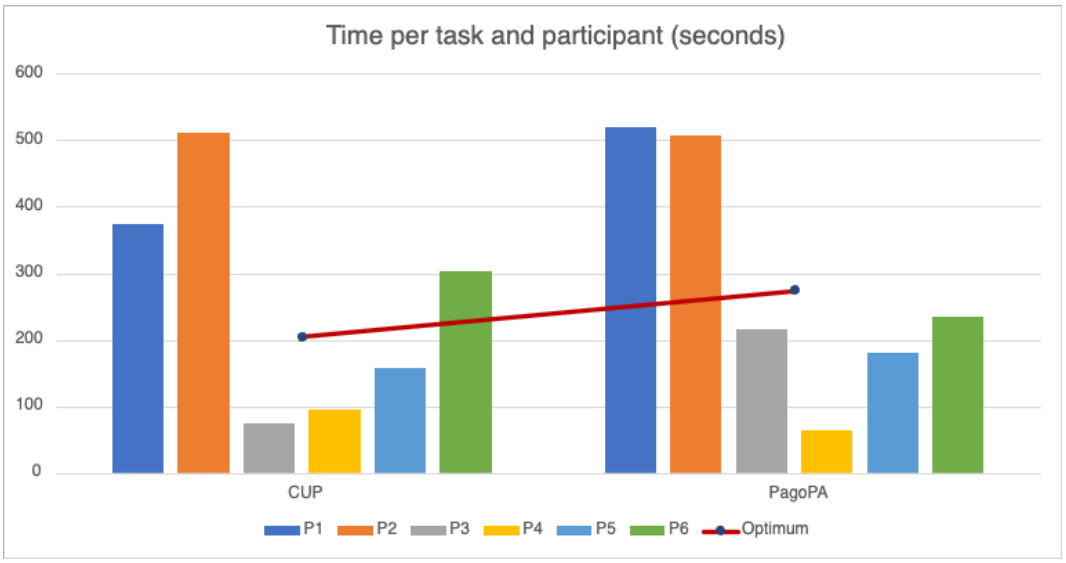


Figure 5. Task completion time per participant (seconds). Dashed lines indicate the optimum time for each task.

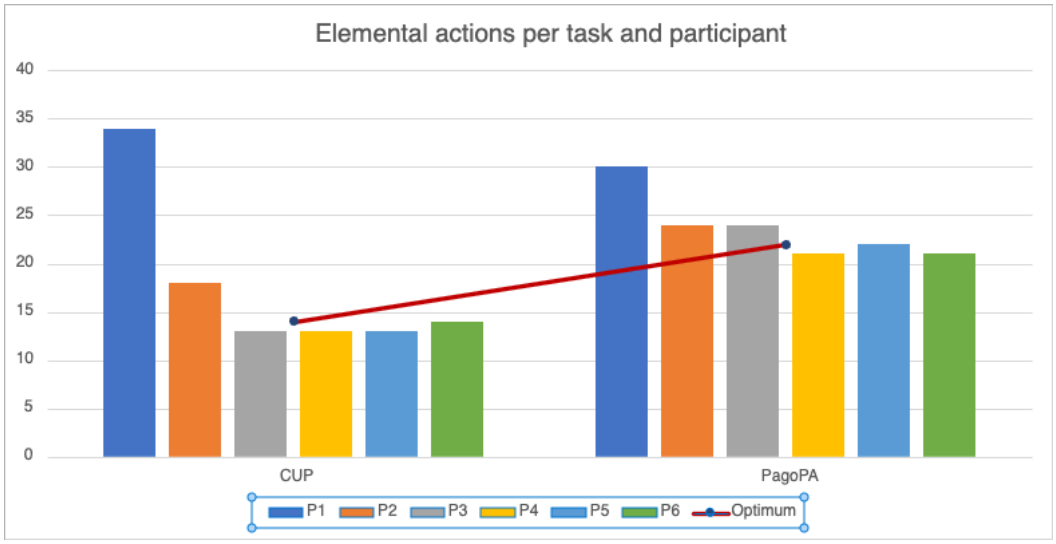


Figure 6. Number of elemental actions per participant and task. Dashed lines indicate the optimum number of actions for each task.

Efficiency findings for the high-fidelity prototype:

- **Finding 1:** Completion time follows a bimodal distribution: P1 and P2 required 373–520 s per task (ratios 1.82–2.52× the optimum), while P3, P4, and P5 completed both tasks well within the optimum and P6 exceeded it only in Task 1 (304 s, ratio 1.48), yielding standard deviations (T1: 175.1 s; T2: 185.7 s) nearly as large as the means themselves.
- **Finding 2:** Action counts are near-optimal for most participants: P3, P4, and P5 completed Task 1 in exactly 13 actions (optimum: 14), while P1 performed 34 actions (a ratio of 2.43×) due to repeated field-correction loops on the CUP form.
- **Finding 3:** For Task 2, only P1 (520 s, ratio 1.89) and P2 (508 s, ratio 1.85) significantly exceed the optimum; the remaining four participants completed the task within the reference optimum, confirming that the pagoPA flow is time-efficient for users with average or higher digital confidence.
- **Finding 4:** P6 completed Task 1 at the optimum action count (14 actions) but required 304 s (ratio 1.48), as a device-specific numeric keyboard was displayed for an alphanumeric input field, requiring facilitator clarification.

5.2 Efficiency analysis

The efficiency comparison between the LoFi and HiFi prototypes can only be made on the basis of elemental actions, as completion time was not measured during the low-fidelity testing phase.

The high-fidelity prototype has comparable efficiency to the low-fidelity prototype on elemental actions, with mixed results across the two tasks.

For Task 1 (CUP Booking), the HiFi mean action count is 17.50 (ratio 1.25) compared to the LoFi Mentor mean of 14.83 (ratio 1.14 against an optimum of 13). The HiFi ratio is slightly higher, primarily driven by P1's 34-action outlier path. The median HiFi action count for Task 1 is 13.5 — close to the LoFi optimum of 13 — indicating that the majority of HiFi participants performed at near-optimal efficiency despite the higher mean being inflated by P1's outlier.

For Task 2 (pagoPA Payment), the HiFi produces a lower ratio (1.08, mean 23.67 actions against an optimum of 22) than the LoFi (1.15, mean 24.17 actions against an optimum of 21), indicating a slight improvement in action efficiency for the payment flow.

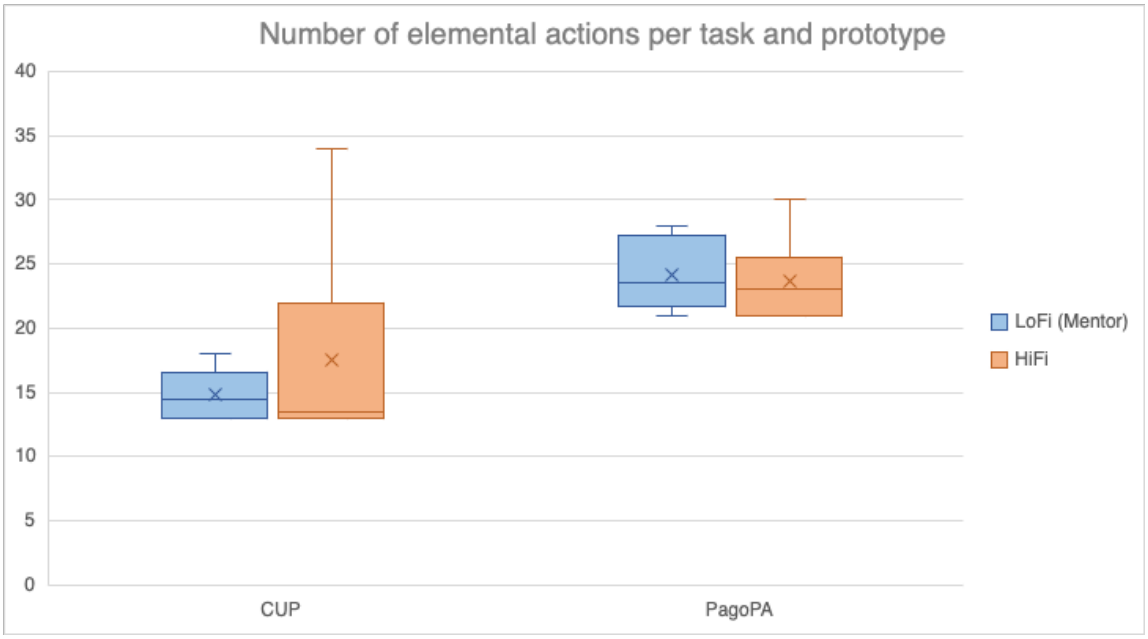


Figure 6b. Box-and-whiskers comparison of elemental actions per task: LoFi Mentor vs HiFi Mentor.

The box-and-whiskers comparison for Task 1 illustrates the key distributional difference: the LoFi action count is tightly clustered around its mean (SD = 1.94), while the HiFi shows a much wider spread (SD = 8.31) driven by P1's outlier. This difference reflects the nature of the evaluation medium rather than a design regression: in the LoFi test, interaction with a paper prototype required only tapping predefined elements, whereas the HiFi requires users to type real data into form fields, an activity that generates additional interactions when users make and correct data-entry mistakes. For Task 2, both prototypes show similarly tight distributions, with the HiFi performing comparably or marginally better. Overall, the HiFi prototype maintains the efficiency profile of the LoFi Mentor for the majority of users, with the additional overhead attributable to the inherently more demanding nature of real form-field interaction.

Efficiency problems of the high-fidelity prototype:

- **Problem 1:** CUP form data entry under cognitive load: insufficient format guidance on the Codice Fiscale and other form fields led P1 to perform 34 actions in Task 1 (2.43× optimum) through repeated correction loops, significantly reducing efficiency for low-confidence users.
- **Problem 2:** Keyboard type mismatch: on certain device models, the prototype displayed a numeric keyboard for fields requiring alphanumeric input (observed in P6's Task 1 session). While the facilitator resolved this with a brief clarification, the issue should be addressed at the prototype level through explicit keyboard type configuration.

6 User satisfaction: SUS questionnaire

6.1 SUS results

SUS score: 78.75 (Standard deviation: 12.32).

Individual scores: P1 = 82.5, P2 = 65.0, P3 = 82.5, P4 = 65.0, P5 = 80.0, P6 = 97.5.

Table 6. SUS response distribution per item (n = 6).

Strongly disagree	Disagree	Neutral	Agree	Strongly agree
1	2	3	4	5

1	0	0	1	3	2
2	2	2	2	0	0
3	0	0	0	4	2
4	3	2	1	0	0
5	0	0	2	2	2
6	3	3	0	0	0
7	0	0	1	3	2
8	2	2	2	0	0
9	0	1	1	2	2
10	2	3	1	0	0

SUS findings for the high-fidelity prototype:

- **Finding 1:** P2 and P4 are the two lowest-scoring participants and drive the main points of negative feedback**.** Both P2 (65.0, 82 years old) and P4 (65.0, 67 years old) scored exactly at the threshold traditionally classified as "OK" (below 68). P2's score is driven primarily by neutral responses (3) on items 2 (unnecessarily complex), 4 (need technical support), and 9 (felt confident), reflecting a persistent perception that the system still requires some degree of assistance for the most elderly participant in the sample. P4's score similarly reflects neutral responses on items 2 and 8 (cumbersome), suggesting that the most technically competent male participants without high digital confidence found the system usable but not friction-free.
- **Finding 2:** Item 4 (need for technical support) receives the most mixed distribution among positive-direction items. 3 participants strongly disagree (1) and 2 disagree (2) that they would need technical support (a strongly positive result) but 1 participant (P2) is neutral (3). This single neutral response from the oldest participant in the sample is structurally meaningful: it indicates that the dual-screen assistant reduced but did not fully eliminate the perceived need for external help for the most vulnerable user in the target population. Combined with P2's observed behaviours (calm but slow, with errors on incomplete form fields), this finding reinforces the priority of improving form field guidance as a means of increasing perceived independence.
- **Finding 3:** SUS item 3 ("I thought the system was easy to use") shows the most mixed distribution among positive items: 1 out of 6 participants responded neutrally, aligning with the form field friction observed during Task 1 and suggesting that the complexity of mandatory data entry reduces perceived ease of use for a subset of the target population.
- **Finding 4:** P6 achieves the highest individual SUS score (97.5), confirming the prototype's strong ceiling performance for digitally familiar users. P6 chose 5 (strongly agree) on all five positive items and 1 (strongly disagree) on all five negative items, yielding a near-perfect score. This is consistent with P6's observed session behaviour: zero mistakes in Task 2, confident use of the left screen for confirmation only, and no facilitator interventions. The 97.5 score from a 64-year-old participant with moderate digital confidence (self-reported 3/5) and partial CUP autonomy is a strong signal that the prototype's usability ceiling is very high even for mid-range users in the target population, and that the lower aggregate mean is driven primarily by two outlier cases rather than by a systemic design problem.

6.2 Analysis of SUS

The SUS score of the high-fidelity prototype is **good**: 78.75 is above the established population average of 68, falling within the "Good" rating band of the standard SUS adjective scale (72.5–85.5). The score indicates that the prototype is perceived as satisfactorily usable by the majority of the target population.

SUS score for the low-fidelity prototype: **95.0** (SD = 4.18).

The high-fidelity prototype has a **lower** SUS score than the low-fidelity prototype, because the two evaluations differ in interaction medium and sample composition. In the LoFi phase, participants interacted with paper mockups that did not require entering real data (e.g., Codice Fiscale, card details), removing the primary friction sources observed in the HiFi sessions. Additionally, the HiFi sample includes P1 and P2 — participants with lower digital confidence than the LoFi group — whose individual scores of 65.0 lower the aggregate mean. The lower HiFi score should therefore be interpreted as a realistic baseline reflecting real-world interaction demands, not as a design regression.

SUS problems of the high-fidelity prototype:

- **Problem 1:** Learnability gap for the most elderly users: the neutral response on item 4 (need for technical support) from P2 (82 years old) indicates that the dual-screen assistant does not fully resolve the perceived dependency on external support for the oldest participant in the sample.
- **Problem 2:** Confidence signal deficit: item 9 (felt confident) received two neutral responses (P2 and P5), suggesting that the prototype does not provide sufficient in-flow confirmation of correct task progress for users who proceed slowly or are uncertain about data entry.

7 User experience: UEQ questionnaire

7.1 UEQ results

The UEQ was completed by all 6 participants. Scale means are reported on the standard −3 to +3 scale. Values above +0.8 indicate a positive evaluation; values in the range −0.8 to +0.8 are neutral; values below −0.8 are negative.

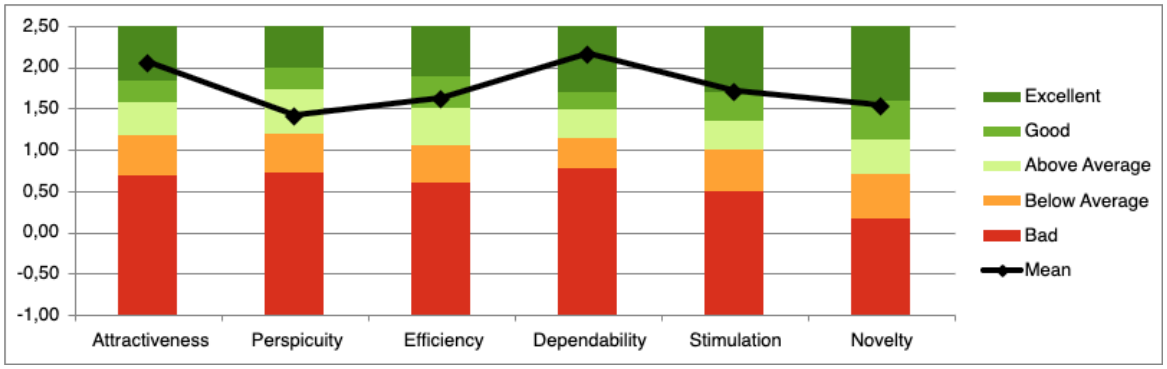


Figure 7. UEQ Benchmark results (n = 6). Scale means are compared against the UEQ benchmark database (21,175 respondents, 468 studies).

UEQ findings for the high-fidelity prototype:

- **Finding 1:** Dependability is the strongest scale (2.17, Excellent) and is anchored by the highest single item in the entire dataset. Item 17 (secure → not secure) has a mean of 2.83 — the highest of all 26 items — and item 11 (obstructive → supportive) scores 2.33. Together, these indicate that users feel strongly in control, safe from errors, and supported throughout their interaction. The linear, step-by-step flow of both task paths, combined with the assistant's explicit on-screen guidance, creates a highly predictable interaction environment that eliminates the uncertainty typically associated with government digital services. Item 19 (meets expectations) at 2.0 further confirms that the prototype consistently matches what users anticipate at each step.
- **Finding 2:** Attractiveness (2.06, Excellent) and Stimulation (1.71, Excellent) reflect strong positive emotional engagement with the prototype. For Attractiveness, the item-level data reveals that users rate the prototype as consistently good (item 12: 2.67), pleasant (item 16: 2.33), friendly (item 25: 2.33), and enjoyable (item 1: 1.83). The only

relatively weaker Attractiveness item is item 24 (attractive → unattractive: 1.33), suggesting that the visual design, while liked overall, has room for greater aesthetic distinctiveness. For Stimulation, item 18 (motivating: 2.17) and item 7 (interesting: 1.83) are both high, while item 6 (boring → exciting: 1.33) and item 5 (valuable → inferior: 1.50) are positive but more moderate, indicating that the intrinsic motivation to use the prototype is strong, though the task domain (medical booking and bill payment) limits its inherent excitement.

- **Finding 3:** Perspicuity is the lowest scale (1.42, Above Average), with item 13 (complicated→easy) as the single weakest item in the dataset (1.00). While all four Perspicuity items are positive, item 13 is the clearest quantitative signal of the learnability friction identified throughout the evaluation. Items 2 (understandable: 1.67) and 21 (clear: 1.50) are moderately positive, while item 4 (easy to learn: 1.50) is also in the positive range. The convergence of the lowest Perspicuity score with the highest concentration of observed task errors (form fields, Codice Fiscale, email mismatch) and the lowest SUS item score (item 13) provides a consistent, triangulated picture of the prototype's one area of genuine friction: the user's ability to intuit the correct data format and input behaviour for mandatory form fields without explicit guidance.

7.2 Analysis of UEQ

Analysis of the six high-level UEQ scales:

- **Attractiveness:** The high-fidelity prototype obtains an excellent result (2.06) because users consistently respond positively to the overall experience. The dual-screen setup, combined with the illustrated assistant character, creates a product identity that is perceived as friendly, good, and enjoyable across the participant sample. Even participants who struggled with specific interaction steps (P1, P2) did not penalise the global impression of the product.
- **Perspicuity:** The high-fidelity prototype obtains an above average result (1.42) because while the majority of users find the system understandable and clear, a subset of participants, particularly those with lower digital confidence, experience friction when confronted with mandatory data entry fields that lack explicit format guidance. Item 13 (complicated→easy: 1.00) is the lowest-scoring item in the entire dataset and directly maps to the Codice Fiscale input challenge in the CUP Booking task. Perspicuity represents the prototype's single area where targeted design improvement would yield the most measurable gain.
- **Efficiency:** The high-fidelity prototype obtains a good result (1.63) because users generally feel they can complete tasks without unnecessary effort. Item 9 (fast→slow: 2.17) scores particularly well, indicating users feel the system responds quickly to their inputs. The more moderate score on item 22 (impractical→practical: 1.00) reflects the real-world friction of public-service form completion, which is not fully reducible by design but could be mitigated through autofill support or clearer field completion cues.
- **Dependability:** The high-fidelity prototype obtains an excellent result (2.17) because the structured, guided interaction flow produces a very strong sense of predictability and control. Item 17 (secure: 2.83) is the highest-rated individual item in the dataset, confirming that users never feel at risk of making irreversible mistakes. This is a critical quality for a prototype designed for elderly users interacting with government digital services, where fear of error is a primary barrier to adoption.
- **Stimulation:** The high-fidelity prototype obtains an excellent result (1.71) because the dual-screen format and the interactive step-by-step assistant transform what would otherwise be a bureaucratic task into an engaging interaction. Item 18 (motivating: 2.17) and item 7 (interesting: 1.83) confirm that users are intrinsically motivated to complete the tasks. The somewhat lower score on item 6 (boring→exciting: 1.33) is consistent with the inherent nature of medical booking and bill payment as tasks, rather than with the prototype design itself.

- **Novelty:** The high-fidelity prototype obtains a good result (1.54) because the dual-screen concept is perceived as innovative and differentiated from conventional single-screen mobile applications. Item 26 (conservative→innovative: 2.33) and item 15 (usual→leading edge: 2.00) reflect this perception clearly. The one below-neutral item in the entire UEQ dataset is item 10 (inventive→conventional: 0.67), which while still slightly positive suggests that the individual UI elements, taken in isolation, follow familiar mobile design patterns. The novelty of the prototype therefore resides primarily in the architectural concept (dual-screen coordination) rather than in the visual design language of individual screens.

The high-fidelity prototype has no UEQ problems.

8 General impressions of participants

8.1 What are the main problems you have found while using this prototype?

Findings on main problems

- **Finding 1:** 4 out of 6 participants (67%) cite initial onboarding uncertainty as the main difficulty, specifically the moment of first connecting to the system and choosing a task (P1: "it was all new, easy to feel anxious"; P2: "mainly at the start, I'm not experienced"; P3: "I didn't understand what to do at the start"; P5: "there was some initial uncertainty"). All four qualify this as temporary and self-resolving once the session progressed, indicating that the prototype's learning curve is steep at first contact but rapidly overcome, which makes it a design opportunity rather than a blocking issue.
- **Finding 2:** 2 out of 6 participants (33%) report no problems of any kind (P4: "I did not find any significant problem"; P6: "with the assistant, it was very easy and intuitive"). The fact that two participants with different confidence profiles, P4 (always manages independently) and P6 (sometimes manages, needs help), both report zero problems confirms that the prototype is capable of delivering a completely frictionless experience for a substantial portion of the target population.

8.2 What is the part of the prototype that has been more difficult to understand? Why?

Findings on parts of the prototype more difficult to understand:

- **Finding 1:** 3 out of 6 participants (50%) identify form field interaction and data entry as the most difficult element (P1: "a lot of fields to fill"; P2: "filling the fields, not sure what to write"; P5: "interacting with the keyboard and entering the data"). This finding converges with Effectiveness Problem 1 and the Perspicuity UEQ scale result, providing independent qualitative confirmation that form field guidance is the single highest-priority design improvement across all data sources collected.
- **Finding 2:** 1 out of 6 participants (17%) found the hospital selection step in Task 1 confusing (P4: "I was not so sure so I felt a bit confused"). This is a content challenge rather than an interface design problem, as the step requires the user to choose among real facilities and time slots without prior contextual knowledge, and is worth noting as a potential area for additional guidance.
- **Finding 3:** 2 out of 6 participants (33%) report nothing was difficult to understand (P3: "there was nothing complex"; P6: "everything very clear and smooth"), confirming that for digitally confident users the prototype's interaction model is fully transparent and requires no learning effort beyond the initial onboarding.

8.3 What have you liked most of the prototype? Why?

Findings on what participants liked most:

- **Finding 1:** 3 out of 6 participants (50%) explicitly identify the step-by-step guided structure as the primary positive element (P2: "it leads you to complete what you have to do"; P3: "very easy to use and intuitive"; P5: "the assistant guides you step-by-step"). This is significant because these three participants span a range of digital confidence levels (2, 3, and 5), suggesting that the value of the guided structure is perceived consistently across the target population and not only by the least experienced users.
- **Finding 2:** P6 highlights the sense of personal autonomy enabled by the prototype: "I found it rewarding to do it on my own without anyone's help." This is the most design-relevant response in the dataset, as it confirms that the Interactive Mentor successfully delivers on its core objective of enabling independent task completion for elderly users who currently rely on others.

8.4 Can you describe your overall experience with this prototype?

Findings on overall experience:

- **Finding 1:** All 6 participants (100%) describe the overall experience in positive terms, ranging from concise endorsements (P1 and P5: "good"; P4: "satisfying") to more articulate assessments (P3: "overall intuitive"; P2: "useful because you have to do it"). The absence of any neutral or negative overall impression, even from P1 and P2 who had the highest error counts, indicates that the prototype's usability problems do not undermine the global user experience and that the design succeeds in creating a positive interaction even for its most challenged users.
- **Finding 2:** P6's response directly validates the prototype's design goal: "it gave me an extra boost to be independent, without the need to go anywhere or rely on others (offices, friends, relatives)." The spontaneous reference to removing dependency on family members and physical offices confirms that the prototype communicates its intended value proposition to users without any facilitation.

8.5 Did you find the two-screen setup clear and easy to follow? Was there anything confusing about it?

Findings on the two-screen setup:

- **Finding 1:** 5 out of 6 participants (83%) describe the two-screen setup in positive terms; only P1 gives a neutral response ("it was not a problem for me"). The breadth of positive endorsement across participants of different ages, confidence levels, and performance profiles confirms that the dual-screen concept has genuine cross-profile acceptance within the target population, and that the initial learning curve associated with it does not translate into lasting confusion.
- **Finding 2**: 3 out of 6 participants independently associate the dual-screen design with reassurance and reduced anxiety around the task (P2: "you are supported"; P4: "helpful in getting tasks done, which often stress me"; P6: "gives me reassurance and peace of mind"), without being prompted on this dimension. The independent convergence on the emotional benefit of the design is particularly strong evidence that the dual-screen architecture addresses a real psychological barrier that the target population faces when approaching mandatory digital services.

9 Summary of usability problems

The table below lists the usability problems identified in the high-fidelity prototype, ordered by relevance.

Table 7. Usability problems of the high-fidelity prototype, ordered by relevance.

High-fidelity prototype	Problem	Severity (cosmetic, minor, major, catastrophe)
Problem 1	The CUP Booking form does not provide format hints, character examples, or inline validation for mandatory fields. The Codice Fiscale field in particular gives no indication of the expected format, and the form does not prevent the user from attempting to advance through incomplete fields. This caused 8 mistakes across P1 and P2 in Task 1 and is the most consistently reported difficulty across all data sources collected.	Major
Problem 2	In the pagoPA flow, an incorrect email entry produces an error state that the user cannot resolve independently. P1 was unable to continue without facilitator intervention, which would not be available in a real-world usage scenario.	Major
Problem 3	4 out of 6 participants (67%) experienced uncertainty at the start of the session when first connecting to the system and selecting a task. The dual-screen setup does not provide a clear initial cue indicating which screen requires the first interaction.	Minor
Problem 4	The payment method selection in the pagoPA flow lacks sufficient visual differentiation between options. 2 out of 6 participants (33%) selected the wrong method before correcting themselves.	Minor
Problem 5	The hospital selection step in the CUP Booking task does not provide contextual information to support the choice (e.g., distance, availability summary). 1 participant (P4) reported uncertainty at this step, and Screen Neglect was observed in P2 specifically during hospital selection.	Minor
Problem 6	The prototype does not provide explicit confirmation feedback between completed steps. 2 out of 6 participants rated "felt very confident using the system" (SUS item 9) at the neutral level, suggesting that the absence of micro-confirmations leaves some users uncertain whether their actions were correct.	Cosmetic

10 Proposals to improve the prototype

Table 8. Proposed improvements to the high-fidelity prototype.

Improvement	Problem(s) addressed	Description of improvement
Improvement 1	Problems 1, 6	Add inline validation to all mandatory fields in the CUP Booking flow, providing real-time format hints, character counters, and positive confirmation (e.g., a visual indicator when a field is correctly filled). This addresses the form field guidance gap and simultaneously provides the in-flow confidence signal missing from the current design.

Improvement 2	Problems 2	Introduce a real-time email validator in the pagoPA flow that detects mismatches immediately upon entry and displays a clear, actionable correction message visible on both the assistant screen and the task screen.
Improvement 3	Problems 3	Add a brief animated orientation cue when the main hub appears for the first time, accompanied by an assistant message indicating which screen requires the first interaction. For repeat users, this cue should be suppressible after the first session.
Improvement 4	Problems 4	Redesign the payment method options with distinct icons and short descriptive labels. Pre-select the most common option by default and require an explicit confirmation tap to reduce accidental selections.
Improvement 5	Problems 5	Display a brief contextual summary alongside each hospital option (e.g., estimated distance or nearest available date) to support an informed selection without requiring prior knowledge of local facilities.
Improvement 6	Problems 3, 6	Add micro-confirmation messages from the assistant at the end of each successfully completed step, starting from the initial task selection at the hub. Applied from the first interaction, this reinforces correct behaviour throughout the flow and reduces the initial orientation uncertainty identified in Problem 3.

11 General conclusions

Team conclusions.

This report concludes a semester-long project that followed the full arc of a user-centred design process, from context of use analysis through two rounds of prototype evaluation. The work began with an investigation into how elderly adults in Italy interact with mandatory digital public services, specifically CUP medical booking and pagoPA bill payment. The context of use analysis surfaced a set of persistent behavioural patterns, including the Hybrid Workaround, Mnemonic Silo, and Safe Haven Search, which shaped the design direction toward a guided, assistive system rather than a self-directed application.

The low-fidelity evaluation compared two competing designs, the Digital Gym and the Interactive Mentor. The Mentor outperformed the Gym on all measured dimensions, producing fewer mistakes, better efficiency ratios, and a substantially higher SUS score (95.0 vs. 70.8). The team selected the Interactive Mentor as the basis for the high-fidelity prototype, iterating on the payment method icons and form field structure based on the LoFi findings.

The high-fidelity evaluation confirmed that the core design proposition holds under realistic interaction conditions. All 6 participants completed both tasks successfully, the SUS score of 78.75 reflects good overall usability, and five of six UEQ scales are rated Good or Excellent. The spontaneous responses collected in the general impressions questionnaire, particularly P6's statement about gaining independence from family and offices, provide qualitative validation of the design goal that motivated the project from the outset.

The main difficulties encountered were the recruitment of participants within the 70-85+ target age range, the logistical complexity of conducting all sessions remotely, and the correction of miscalculated optimal values from the planning document. These are acknowledged as limitations of the current evaluation cycle and do not affect the validity of the findings.

Personal conclusions.

This project gave us a direct, concrete experience of what it means to design for a population whose needs are rarely centered in mainstream product development. The gap between a design that seems accessible and one that genuinely is accessible only became visible through repeated contact with real users from the target group. The moments that surprised us most, such as watching a participant spend several minutes on a task we had estimated would take under two, or hearing someone describe an experience as rewarding because they completed it alone for the first time, were also the ones that taught us the most. We leave this project with a clearer understanding of how usability evaluation works not as a final checkpoint but as an iterative tool, and with a genuine appreciation for the role that methodological rigour plays in producing results that are worth trusting.

12 Credits

Full name	Activities performed	% of work done
Cirillo, Riccardo	Facilitator for Sessions 5 and 6 (P5, P6); observer for Sessions 1 and 2 (P1, P2); quantitative data analysis (effectiveness and efficiency metrics); report writing (sections 5 and 6).	25%
Papapicco, Fabio Vito	Facilitator for Sessions 1, 2, 3, and 4 (P1, P2, P3, P4); participant recruitment and session logistics; HiFi prototype refinement; data preparation and chart generation; report writing (sections 2, 3, and 4).	25%
Ramaglia Bianchino, Andrea	Observer for Sessions 1, 4, 5, and 6 (P1, P4, P5, P6); SUS and UEQ data processing and analysis; report writing (sections 7, 10, 11, and 12).	25%
Salami, Ilaria	Observer for Sessions 2, 3, 4, 5, and 6 (P2, P3, P4, P5, P6); qualitative data analysis (observation sheets and general impression questionnaires); report writing (sections 8 and 9); report review and final editing.	25%

and the processing of the data, in addition to the periods established in the regulations on archives and documentation.

What is the legitimacy for the processing of your data?

The legal bases that legitimize the processing are those set out in the General Data Protection Regulation (GDPR) of the European Union (EU) 2016/679:

- Article 6.1.a) the data subject has given his/her consent to the processing of his/her personal data for one or more specific purposes.
- Article 6.1.e) "Processing necessary for the fulfilment of a mission of public interest", consisting of a practice that is carried out with a learning objective.

Are you obliged to provide this personal data and what are the consequences of not doing so?

The data requested are those that are strictly necessary to carry out the practical activity, which is why it will not be possible for you to participate if you do not provide them, due to the totally voluntary nature of this learning project.

To whom will your data be communicated?

- There are no plans to communicate your data to third parties and no international transfers will be made.
- The identity of the participants will not be revealed at any time. Only students who carry out the practical activity will be aware of the personal data of each participant.

What are your rights in relation to the data provided?

- In relation to the data provided, you have the right to access, rectify, request cancellation and deletion and oppose or limit the processing of the same, under the terms established by law.
- You can obtain more information on the exercise of these rights by contacting the student acting as Data Protection Officer.
- Likewise, and if you do not obtain satisfaction in the exercise of your rights, you may file a complaint with the Spanish Data Protection Agency: https://www.aepd.es/es

12.2 Observation Sheets

Observation Sheet

Participant ID: P1 Date: 12/05/2026 Time: 20:40

Facilitator: Fabio Vito Papapicco Observer(s): Riccardo Cirillo, Andrea Ramaglia Bianchino

Intended use: This observation sheet will be used as an electronic document. During the remote session, the observer(s) will fill it in real time directly in a Word file, recording objective data and behavioural observations for each participant and each task.

Task 1 – CUP Booking

Task 1 – Task Execution Log

Measurement	Value	Observer's Comments
Success	[x] Yes [ ] No	Completed successfully with few hesitation

Measurement	Value	Observer's Comments
Completion Time	06 min 13 sec	
Total Actions (Taps/Clicks)	34	
Total Mistakes	5	
Relevant Observations	*(body language, hesitations, physical difficulties)*	
Initial screen orientation uncertainty, read mentor aloud, hesitation during tutorial, used mentor consistently for form, uncertainty on Codice Fiscale data, completed successfully ..		

Task 1 – Behavioral & Support Checklist

Behavior Observed	Yes	No	Specific Details (What did they say/do?)
Hybrid Workaround: Did they use a paper notebook or pen?	[ ]	[x]	
Safe Haven Search: Did they look for reassurance before clicking?	[ ]	[x]	No, although sometimes the participants has to read again the tutorial screen
Mnemonic Freeze: Did they "lose the thread" during a sequence of actions?	[ ]	[x]	
System Avoidance: Did they refuse to click for fear of making mistakes?	[ ]	[x]	
Screen Neglect: Did they skip or ignore the left screen without reading it fully?	[ ]	[x]	Always reading carefully

Task 2 – pagoPA Payment

Task 2 – Task Execution Log

Measurement	Value	Observer's Comments
Success	[x] Yes [ ] No	
Completion Time	08 min 40 sec	
Total Actions (Taps/Clicks)	30	
Total Mistakes	5	
Relevant Observations	*(body language, hesitations, physical difficulties)*	
quick correct path selection, QR doubt but correct execution, careful data checking, email uncertainty + asked facilitator, blocked by mismatch error requiring facilitator help, hesitations on card form scrolling, placeholder used for name field, completed successfully		

Task 2 – Behavioral & Support Checklist

Behavior Observed	Yes	No	Specific Details (What did they say/do?)
Hybrid Workaround: Did they use a paper notebook or pen?	[ ]	[x]	
Safe Haven Search: Did they look for reassurance before clicking?	[x]	[ ]	Specifically when having to fill the email field.
Mnemonic Freeze: Did they "lose the thread" during a sequence of actions?	[ ]	[x]	
System Avoidance: Did they refuse to click for fear of making mistakes?	[ ]	[x]	
Screen Neglect: Did they skip or ignore the left screen without reading it fully?	[ ]	[x]	

Observer's Post-Test Synthesis

Main friction points	CUP: Initial orientation (which screen to start from); uncertainty on Codice Fiscale data; understanding how to use the form. PagoPA: Email entry and confirmation (mismatch not self-corrected, required facilitator intervention); scrolling through card form with hesitations; uncertainty on cardholder name field.
User's confidence level (Observation)	[ ] Very Low [ ] Low [x] Neutral [ ] High [ ] Very High
Unexpected mental models	Participant appeared to expect form fields to have more explicit guidance on the format of data to enter (e.g. Codice Fiscale, cardholder name)
Critical failures (if any)	None. Both tasks completed successfully.

Observation Sheet

Participant ID: P2 Date: 13-5-26 Time: 12:30

Facilitator: FABIO PAPAPICCO Observer(s): ILARIA SALAMI

Intended use: This observation sheet will be used as an electronic document. During the remote session, the observer(s) will fill it in real time directly in a Word file, recording objective data and behavioural observations for each participant and each task.

Task 1 – CUP Booking

Task 1 – Task Execution Log

Measurement	Value	Observer's Comments
Success	[X] Yes [ ] No	no major problem/difficulties
Completion Time	8 min 37 sec	
Total Actions (Taps/Clicks)	18	
Total Mistakes	3	mostly when trying to move on when a field was incomplete.

Relevant Observations	*(body language, hesitations, physical difficulties)*
body language suggests the participant was not stressing and was overall in a calm state, even when facing minor difficulties	

Task 1 – Behavioral & Support Checklist

Behavior Observed	Yes	No	Specific Details (What did they say/do?)
Hybrid Workaround: Did they use a paper notebook or pen?	[ ]	[X]	
Safe Haven Search: Did they look for reassurance before clicking?	[ ]	[X]	Pretty confident, even when taking his time the participant did not look for approval
Mnemonic Freeze: Did they "lose the thread" during a sequence of actions?	[ ]	[X]	
System Avoidance: Did they refuse to click for fear of making mistakes?	[ ]	[X]	never seemed worried or scared
Screen Neglect: Did they skip or ignore the left screen without reading it fully?	[X]	[ ]	yes, especially when having to choose an hospital

Task 2 – pagoPA Payment

Task 2 – Task Execution Log

Measurement	Value	Observer's Comments
Success	[X] Yes [ ] No	no major issue encountered
Completion Time	8 min 28 sec	
Total Actions (Taps/Clicks)	24	
Total Mistakes	3	He did not pick the right payment method
Relevant Observations	*(body language, hesitations, physical difficulties)*	
body language suggests the participant was not stressing and was overall in a calm state, even when facing minor difficulties		

Task 2 – Behavioral & Support Checklist

Behavior Observed	Yes	No	Specific Details (What did they say/do?)
Hybrid Workaround: Did they use a paper notebook or pen?	[ ]	[X]	
Safe Haven Search: Did they look for reassurance before clicking?	[ ]	[X]	Pretty confident, even when taking his time the participant did not look for approval
Mnemonic Freeze: Did they "lose the thread" during a sequence of actions?	[ ]	[X]	always kept his focus
System Avoidance: Did they refuse to click for fear of making mistakes?	[ ]	[X]	confident, never seemed worries
Screen Neglect: Did they skip or ignore the left screen without reading it fully?	[ ]	[X]	

Post-Test Synthesis

Observer's Post-Test Synthesis

Main friction points	when having to fill fields, often wrote something wrong or incomplete. For the rest, no friction points.
User's confidence level (Observation)	[ ] Very Low [ ] Low [ ] Neutral [X] High [ ] Very High
Unexpected mental models	none
Critical failures (if any)	none

Observation Sheet

Participant ID: P3 Date: _ 31/05/2026 Time: 17:30

Facilitator: Fabio Vito Papapicco Observer(s): Ilaria Salami

Intended use: This observation sheet will be used as an electronic document. During the remote session, the observer(s) will fill it in real time directly in a Word file, recording objective data and behavioural observations for each participant and each task.

Task 1 – CUP Booking

Task 1 – Task Execution Log

Measurement	Value	Observer's Comments
Success	[x] Yes [ ] No	No issues and no hesitations
Completion Time	1 min 14 sec	
Total Actions (Taps/Clicks)	13	
Total Mistakes	0	
Relevant Observations	*(body language, hesitations, physical difficulties)*	

Measurement	Value	Observer's Comments
Participant seemed very confident, if needed they were looking at the left screen for confirmation		

Task 1 – Behavioral & Support Checklist

Behavior Observed	Yes	No	Specific Details (What did they say/do?)
Hybrid Workaround: Did they use a paper notebook or pen?	[ ]	[x]	
Safe Haven Search: Did they look for reassurance before clicking?	[ ]	[x]	
Mnemonic Freeze: Did they "lose the thread" during a sequence of actions?	[ ]	[x]	
System Avoidance: Did they refuse to click for fear of making mistakes?	[ ]	[x]	
Screen Neglect: Did they skip or ignore the left screen without reading it fully?	[ ]	[x]	For most clicks they were using the left screen as confirmation on what to do. Regardless the participant never seemed lost

Task 2 – pagoPA Payment

Task 2 – Task Execution Log

Measurement	Value	Observer's Comments
Success	[x] Yes [ ] No	Succeeded with no issue
Completion Time	3 min 35 sec	

Measurement	Value	Observer's Comments
Total Actions (Taps/Clicks)	24	
Total Mistakes	2	Wrong payment method selected
Relevant Observations	*(body language, hesitations, physical difficulties)*	
Pretty confident, no issue encountered, slight hesitation on having to add card's number field		

Task 2 – Behavioral & Support Checklist

Behavior Observed	Yes	No	Specific Details (What did they say/do?)
Hybrid Workaround: Did they use a paper notebook or pen?	[ ]	[x]	
Safe Haven Search: Did they look for reassurance before clicking?	[ ]	[x]	
Mnemonic Freeze: Did they "lose the thread" during a sequence of actions?	[ ]	[x]	
System Avoidance: Did they refuse to click for fear of making mistakes?	[ ]	[x]	
Screen Neglect: Did they skip or ignore the left screen without reading it fully?	[x]	[ ]	Participant moved confidently within the prototype, mostly did not have to look at the left screen

Post-Test Synthesis

Observer's Post-Test Synthesis

Main friction points	Insertion of payment card information
User's confidence level (Observation)	[ ] Very Low [ ] Low [ ] Neutral [x] High [ ] Very High
Unexpected mental models	Nothing unexpected
Critical failures (if any)	none

Observation Sheet

Participant ID: P4 Date: 31/05/2026 Time: 17.45

Facilitator: Fabio Vito Papapicco Observer(s): Ilaria Salami, Andrea Ramaglia Bianchino

Intended use: This observation sheet will be used as an electronic document. During the remote session, the observer(s) will fill it in real time directly in a Word file, recording objective data and behavioural observations for each participant and each task.

Task 1 – CUP Booking

Task 1 – Task Execution Log

Measurement	Value	Observer's Comments
Success	[x] Yes [ ] No	
Completion Time	01 min 34 sec	
Total Actions (Taps/Clicks)	13	
Total Mistakes	0	
Relevant Observations	*(body language, hesitations, physical difficulties)*	
condident, no hesitation		

Task 1 – Behavioral & Support Checklist

Behavior Observed	Yes	No	Specific Details (What did they say/do?)
Hybrid Workaround: Did they use a paper notebook or pen?	[ ]	[x]	
Safe Haven Search: Did they look for reassurance before clicking?	[ ]	[x]	
Mnemonic Freeze: Did they "lose the thread" during a sequence of actions?	[ ]	[x]	
System Avoidance: Did they refuse to click for fear of making mistakes?	[]	[x]	
Screen Neglect: Did they skip or ignore the left screen without reading it fully?	[x]	[ ]	participant barely needed the instructions on the left screen

Task 2 – pagoPA Payment

Task 2 – Task Execution Log

Measurement	Value	Observer's Comments
Success	[x] Yes [ ] No	
Completion Time	01 min 03 sec	
Total Actions (Taps/Clicks)	21	
Total Mistakes	0	
Relevant Observations	*(body language, hesitations, physical difficulties)*	
none, participant was confident and did not hesitate		

Task 2 – Behavioral & Support Checklist

Behavior Observed	Yes	No	Specific Details (What did they say/do?)
Hybrid Workaround: Did they use a paper notebook or pen?	[ ]	[x]	
Safe Haven Search: Did they look for reassurance before clicking?	[ ]	[x]	
Mnemonic Freeze: Did they "lose the thread" during a sequence of actions?	[ ]	[x]	
System Avoidance: Did they refuse to click for fear of making mistakes?	[ ]	[x]	
Screen Neglect: Did they skip or ignore the left screen without reading it fully?	[x]	[ ]	participant barely needed the instructions on the left screen

Post-Test Synthesis

Observer's Post-Test Synthesis

Main friction points	none
User's confidence level (Observation)	[ ] Very Low [ ] Low [ ] Neutral [ ] High [x] Very High
Unexpected mental models	none
Critical failures (if any)	none

Observation Sheet

Participant ID: P5 Date: 01/06/2026 Time: 12:30

Facilitator: Riccardo Cirillo Observer(s): Ilaria Salami, Andrea Ramaglia Bianchino

Intended use: This observation sheet will be used as an electronic document. During the remote session, the observer(s) will fill it in real time directly in a Word file, recording objective data and behavioural observations for each participant and each task.

Task 1 – CUP Booking

Task 1 – Task Execution Log

Measurement	Value	Observer's Comments
Success	[x] Yes [ ] No	Completed successfully with few hesitation
Completion Time	02 min 37 sec	
Total Actions (Taps/Clicks)	13	
Total Mistakes	0	
Relevant Observations	*(body language, hesitations, physical difficulties)*	
Initial screen orientation uncertainty, read mentor aloud, hesitation during tutorial, used mentor consistently for form, uncertainty on Codice Fiscale data, completed successfully ..		

Task 1 – Behavioral & Support Checklist

Behavior Observed	Yes	No	Specific Details (What did they say/do?)
Hybrid Workaround: Did they use a paper notebook or pen?	[ ]	[x]	
Safe Haven Search: Did they look for reassurance before clicking?	[ ]	[x]	
Mnemonic Freeze: Did they "lose the thread" during a sequence of actions?	[ ]	[x]	

Behavior Observed	Yes	No	Specific Details (What did they say/do?)
System Avoidance: Did they refuse to click for fear of making mistakes?	[ ]	[x]	
Screen Neglect: Did they skip or ignore the left screen without reading it fully?	[x]	[ ]	Hardly ever needed left screen

Task 2 – pagoPA Payment

Task 2 – Task Execution Log

Measurement	Value	Observer's Comments
Success	[x] Yes [ ] No	
Completion Time	03 min 01 sec	
Total Actions (Taps/Clicks)	22	
Total Mistakes	1	quite irrelevant, momentarily hides the keyboard when they still needed it
Relevant Observations	*(body language, hesitations, physical difficulties)*	
participant took their time, was not super confident but proceeded without problems		

Task 2 – Behavioral & Support Checklist

Behavior Observed	Yes	No	Specific Details (What did they say/do?)
Hybrid Workaround: Did they use a paper notebook or pen?	[ ]	[x]	
Safe Haven Search: Did they look for reassurance before clicking?	[ ]	[x]	

Behavior Observed	Yes	No	Specific Details (What did they say/do?)
Mnemonic Freeze: Did they "lose the thread" during a sequence of actions?	[ ]	[x]	
System Avoidance: Did they refuse to click for fear of making mistakes?	[ ]	[x]	
Screen Neglect: Did they skip or ignore the left screen without reading it fully?	[ ]	[x]	used it for support

Post-Test Synthesis

Observer's Post-Test Synthesis

Main friction points	None
User's confidence level (Observation)	[ ] Very Low [ ] Low [ ] Neutral [x] High [ ] Very High
Unexpected mental models	None
Critical failures (if any)	None.

Observation Sheet

Participant ID: P6 Date: 01/06/2026 Time: 14:00

Facilitator: Riccardo Cirillo Observer(s): Ilaria Salami, Andrea Ramaglia

Intended use: This observation sheet will be used as an electronic document. During the remote session, the observer(s) will fill it in real time directly in a Word file, recording objective data and behavioural observations for each participant and each task.

Task 1 – CUP Booking

Task 1 – Task Execution Log

Measurement	Value	Observer's Comments
Success	[x] Yes [ ] No	
Completion Time	5 min 4 sec	
Total Actions (Taps/Clicks)	14	
Total Mistakes	1	Hid the keyboard even when not done filling the field
Relevant Observations	*(body language, hesitations, physical difficulties)*	
No major issue, took his time filling the fields especially cause given the model of his device he was shown a numeric keyboard when needing alphanumeric. The facilitator intervened just to say he could use only numbers and it would be fine.		

Task 1 – Behavioral & Support Checklist

Behavior Observed	Yes	No	Specific Details (What did they say/do?)
Hybrid Workaround: Did they use a paper notebook or pen?	[ ]	[x]	
Safe Haven Search: Did they look for reassurance before clicking?	[ ]	[x]	
Mnemonic Freeze: Did they "lose the thread" during a sequence of actions?	[ ]	[x]	
System Avoidance: Did they refuse to click for fear of making mistakes?	[ ]	[x]	
Screen Neglect: Did they skip or ignore the left screen without reading it fully?	[ ]	[x]	Looked at the left screen for reassurance

Task 2 – pagoPA Payment

Task 2 – Task Execution Log

Measurement	Value	Observer's Comments
Success	[x] Yes [ ] No	
Completion Time	3 min 55 sec	
Total Actions (Taps/Clicks)	21	
Total Mistakes	0	
Relevant Observations	*(body language, hesitations, physical difficulties)*	
No mistakes, pretty confident.		

Task 2 – Behavioral & Support Checklist

Behavior Observed	Yes	No	Specific Details (What did they say/do?)
Hybrid Workaround: Did they use a paper notebook or pen?	[ ]	[x]	
Safe Haven Search: Did they look for reassurance before clicking?	[ ]	[x]	
Mnemonic Freeze: Did they "lose the thread" during a sequence of actions?	[ ]	[x]	
System Avoidance: Did they refuse to click for fear of making mistakes?	[ ]	[x]	
Screen Neglect: Did they skip or ignore the left screen without reading it fully?	[ ]	[x]	Looked at the left screen to make sure to do the right thing

Post-Test Synthesis

Observer's Post-Test Synthesis

Main friction points	None
User's confidence level (Observation)	[ ] Very Low [ ] Low [ ] Neutral [x] High [ ] Very High
Unexpected mental models	none
Critical failures (if any)	none

12.3 General Impressions

General Impression

Participant ID	P1
Date and time	12/05/2026 21:00
1. What are the main problems you have found while using this prototype?	
At the start some indecisiveness, it was all new so it is easy to feel anxious but the instructions helped.	
2. What is the part of the prototype that has been more difficult to understand? Why?	
Task 1, there were a lot of fields to fill.	
3. What have you liked most of the prototype? Why?	
I wouldn't know, everything overall.	
4. Can you describe your overall experience with this prototype?	
Good.	
5. Did you find the two-screen setup clear and easy to follow? Was there anything confusing about it?	
It was not a problem for me	

General Impression

Participant ID	P2
Date and time	12:30

13·05·2026,

1. What are the main problems you have found while using this prototype?
Mainly at the start since Im not experienced with this.
2. What is the part of the prototype that has been more difficult to understand? Why?
specifically filling the fields cause Im not sure what to write
3. What have you liked most of the prototype? Why?
nothing was easy but its a way to do this task that lead you to complete what you have to do at least
4. Can you describe your overall experience with this prototype?
It is usefulbecause you have to do it
5. Did you find the two-screen setup clear and easy to follow? Was there anything confusing about it?
It's a good solution since you are supported

General Impression

Participant ID	P3
Date and time	31/05/2026 17:20
1. What are the main problems you have found while using this prototype?	
I didn't really understand what I had to do at the start. Not the system fault but mine. Anyway, I did not face any problem during the tasks, only the start.	
2. What is the part of the prototype that has been more difficult to understand? Why?	
There was nothing complex to understand.	
3. What have you liked most of the prototype? Why?	
How clear it was, I think it is very easy to use and intuitive.	
4. Can you describe your overall experience with this prototype?	
Besides the start, overall intuitive.	
4.1 (follow-up) What exactly at the start?	
The start, when we connect to the system and we have to choose the task.	
5. Did you find the two-screen setup clear and easy to follow? Was there anything confusing about it?	
I had no issue in clarity having the two screens.	

General Impression

Participant ID	P4
Date and time	31/05/2026 - 18:10
1. What are the main problems you have found while using this prototype?	
I did not find any significant problem	
2. What is the part of the prototype that has been more difficult to understand? Why?	
In the choice of the hospital I was not so sure so I felt a bit confused.	
3. What have you liked most of the prototype? Why?	
I liked the PagoPa part, I found it easy and fast to use.	
4. Can you describe your overall experience with this prototype?	
I'd say satisfying.	
5. Did you find the two-screen setup clear and easy to follow? Was there anything confusing about it?	
No no, I'd say very easy and I would add it was helpful in getting this tasks done, which often stress me.	

General Impression

Participant ID	P5
Date and time	01/06/2026 12:30
1. What are the main problems you have found while using this prototype?	
Seeing it for the first time, there was some initial uncertainty.	
2. What is the part of the prototype that has been more difficult to understand? Why?	
Interacting with the keyboard and entering the data.	
3. What have you liked most of the prototype? Why?	
The assistant is good because it guides you step-by-step	
4. Can you describe your overall experience with this prototype?	
Good	
5. Did you find the two-screen setup clear and easy to follow? Was there anything confusing about it?	

After the first few steps, it becomes clear and easy to follow.

General Impression

Participant ID	P6
Date and time	01/06/2026 14:00
1. What are the main problems you have found while using this prototype?	
With the assistant, it was very easy and intuitive.	
2. What is the part of the prototype that has been more difficult to understand? Why?	
I didn't find any; everything seems very clear and smooth.	
3. What have you liked most of the prototype? Why?	
As someone who doesn't use technology very much, I found it rewarding to do it on my own without anyone's help	
4. Can you describe your overall experience with this prototype?	
It gave me an extra boost to be independent, without the need to go anywhere or rely on others (offices, friends, relatives, etc.)	
5. Did you find the two-screen setup clear and easy to follow? Was there anything confusing about it?	
Very functional; the dual screen gives me reassurance and peace of mind by guiding me step-by-step.	

12.4 Personal Information Questionnaire

Personal Information Questionnarie

Questionnaire:

Participant ID: P1

Date: 12/05/2026

Time: 20:38

Age: 74

Gender: [ ] M [x] F [ ] Other

Digital Habits:

Average daily time using a personal computer:

[x] Never [ ] <1h [ ] 1–3h [ ] >3h

Average daily time using a smartphone:

[ ] Never [ ] <1h [x] 1–3h [ ] >3h

On a scale of 1 to 5, how confident do you feel using digital services on your own?

(1 = Not at all, 5 = Very confident)

Score: [2]

Questions of Interest:

Do you usually use a paper notebook to write down passwords or steps to follow?

[x] Yes [ ] No

When you encounter a new app or interface, what is your first reaction?

[ ] Curiosity [ ] Fear of breaking something [x] Need to ask for help

Do you usually manage your medical bookings (CUP) or bill payments (pagoPA) independently?

[ ] Yes, always [ ] Sometimes [x] No, I delegate to others

Personal Information Questionnarie

Questionnaire:

Participant ID: P2

Date: 13/05/2026

Time: 12:30

Age: 82

Gender: [x] M [ ] F [ ] Other

Digital Habits:

Average daily time using a personal computer:

[ ] Never [ ] <1h [x] 1–3h [ ] >3h

Average daily time using a smartphone:

[ ] Never [x] <1h [ ] 1–3h [ ] >3h

On a scale of 1 to 5, how confident do you feel using digital services on your own?

(1 = Not at all, 5 = Very confident)

Score: [3]

Questions of Interest:

Do you usually use a paper notebook to write down passwords or steps to follow?

[x] Yes [ ] No

When you encounter a new app or interface, what is your first reaction?

[x] Curiosity [ ] Fear of breaking something [x] Need to ask for help

Do you usually manage your medical bookings (CUP) or bill payments (pagoPA) independently?

[ ] Yes, always [x] Sometimes [ ] No, I delegate to others

Personal Information Questionnarie

Questionnaire:

Participant ID: P3

Date: 31/05/2026

Time: 17:20

Age: 53

Gender: [ ] M [x] F [ ] Other

Digital Habits:

Average daily time using a personal computer:

[ ] Never [ ] <1h [ ] 1–3h [x] >3h

Average daily time using a smartphone:

[ ] Never [ ] <1h [x] 1–3h [ ] >3h

On a scale of 1 to 5, how confident do you feel using digital services on your own?

(1 = Not at all, 5 = Very confident)

Score: [3]

Questions of Interest:

Do you usually use a paper notebook to write down passwords or steps to follow?

[x] Yes [ ] No

When you encounter a new app or interface, what is your first reaction?

[ ] Curiosity [x] Fear of breaking something [ ] Need to ask for help

Do you usually manage your medical bookings (CUP) or bill payments (pagoPA) independently?

[ ] Yes, always [x] Sometimes [ ] No, I delegate to others

Personal Information Questionnarie

Questionnaire:

Participant ID: P4

Date: 31/05/2026

Time: 17:45

Age: 67

Gender: [x] M [ ] F [ ] Other

Digital Habits:

Average daily time using a personal computer:

[ ] Never [ ] <1h [ ] 1–3h [X] >3h

Average daily time using a smartphone:

[ ] Never [x] <1h [ ] 1–3h [ ] >3h

On a scale of 1 to 5, how confident do you feel using digital services on your own?

(1 = Not at all, 5 = Very confident)

Score: [3]

Questions of Interest:

Do you usually use a paper notebook to write down passwords or steps to follow?

[x] Yes [ ] No

When you encounter a new app or interface, what is your first reaction?

[ ] Curiosity [ ] Fear of breaking something [x] Need to ask for help

Do you usually manage your medical bookings (CUP) or bill payments (pagoPA) independently?

[x] Yes, always [ ] Sometimes [ ] No, I delegate to others

Personal Information Questionnarie

Questionnaire:

Participant ID: P5

Date: 01/06/2026

Time: 12:30

Age: 60

Gender: [ ] M [X] F [ ] Other

Digital Habits:

Average daily time using a personal computer:

[ ] Never [ ] <1h [ ] 1–3h [x] >3h

Average daily time using a smartphone:

[ ] Never [ ] <1h [ ] 1–3h [x] >3h

On a scale of 1 to 5, how confident do you feel using digital services on your own?

(1 = Not at all, 5 = Very confident)

Score: [5]

Questions of Interest:

Do you usually use a paper notebook to write down passwords or steps to follow?

[ ] Yes [x] No

When you encounter a new app or interface, what is your first reaction?

[x] Curiosity [ ] Fear of breaking something [ ] Need to ask for help

Do you usually manage your medical bookings (CUP) or bill payments (pagoPA) independently?

[x] Yes, always [ ] Sometimes [ ] No, I delegate to others

Personal Information Questionnarie

Questionnaire:

Participant ID: P6

Date: 01/06/2026

Time: 14:00

Age: 64

Gender: [X] M [ ] F [ ] Other

Digital Habits:

Average daily time using a personal computer:

[ ] Never [ ] <1h [X] 1–3h [ ] >3h

Average daily time using a smartphone:

[ ] Never [ ] <1h [X] 1–3h [ ] >3h

On a scale of 1 to 5, how confident do you feel using digital services on your own?

(1 = Not at all, 5 = Very confident)

Score: [3]

Questions of Interest:

Do you usually use a paper notebook to write down passwords or steps to follow?

[X] Yes [ ] No

When you encounter a new app or interface, what is your first reaction?

[ ] Curiosity [ ] Fear of breaking something [X] Need to ask for help

Do you usually manage your medical bookings (CUP) or bill payments (pagoPA) independently?

[ ] Yes, always [X] Sometimes [ ] No, I delegate to others

12.5 Questionnaire SUS

Questionnaire SUS

Participant ID	P1
Date and time	12/05/2026 - 20:59

Reply with your degree of agreement or disagreement to the following ten sentences, where 1 means “I totally disagree with the sentence” and 5 means “I totally agree with the sentence”.

	1	**2**	**3**	**4**	**5**
I think that I would like to use this system frequently.				x	
I found the system unnecessarily complex.			x		
I thought the system was easy to use.					x
I think that I would need the support of a technical person to be able to use this system.		x			
I found the various functions in this system were well integrated.					x
I thought there was too much inconsistency in this system.		x			
I would imagine that most people would learn to use this system very quickly.					x
I found the system very cumbersome to use.	x				
I felt very confident using the system.					x
I needed to learn a lot of things before I could get going with this system.			x		

12.6 Questionnaire SUS

Participant ID	P2

Date and time	13-5-2026 12:30

Reply with your degree of agreement or disagreement to the following ten sentences, where 1 means “I totally disagree with the sentence” and 5 means “I totally agree with the sentence”.

	1	2	3	4	5
I think that I would like to use this system frequently.			●		
I found the system unnecessarily complex.		●			
I thought the system was easy to use.	●				
I think that I would need the support of a technical person to be able to use this system.			●		
I found the various functions in this system were well integrated.			●		
I thought there was too much inconsistency in this system.	●				
I would imagine that most people would learn to use this system very quickly.					●
I found the system very cumbersome to use.			●		
I felt very confident using the system.		●			
I needed to learn a lot of things before I could get going with this system.		●			

Questionnaire SUS

Participant ID	P3
Date and time	31/05/2026 - 17:30

Reply with your degree of agreement or disagreement to the following ten sentences, where 1 means “I totally disagree with the sentence” and 5 means “I totally agree with the sentence”.

	1	2	3	4	5
I think that I would like to use this system frequently.				x	
I found the system unnecessarily complex.	x				
I thought the system was easy to use.				x	

	1	2	3	4	5
I think that I would need the support of a technical person to be able to use this system.	x				
I found the various functions in this system were well integrated.				x	
I thought there was too much inconsistency in this system.		x			
I would imagine that most people would learn to use this system very quickly.				x	
I found the system very cumbersome to use.		x			
I felt very confident using the system.				x	
I needed to learn a lot of things before I could get going with this system.	x				

Questionnaire SUS

Participant ID	P4
Date and time	31/05/2026 - 18:00

Reply with your degree of agreement or disagreement to the following ten sentences, where 1 means “I totally disagree with the sentence” and 5 means “I totally agree with the sentence”.

	1	2	3	4	5
I think that I would like to use this system frequently.				x	
I found the system unnecessarily complex.			x		
I thought the system was easy to use.				x	
I think that I would need the support of a technical person to be able to use this system.		x			
I found the various functions in this system were well integrated.			x		
I thought there was too much inconsistency in this system.		x			
I would imagine that most people would learn to use this system very quickly.			x		
I found the system very cumbersome to use.			x		
I felt very confident using the system.				x	
I needed to learn a lot of things before I could get going with this system.		x			

Questionnaire SUS

Participant ID	P5

Date and time	01/06/2026 12:30

Reply with your degree of agreement or disagreement to the following ten sentences, where 1 means “I totally disagree with the sentence” and 5 means “I totally agree with the sentence”.

	1	2	3	4	5
I think that I would like to use this system frequently.					X
I found the system unnecessarily complex.		X			
I thought the system was easy to use.				X	
I think that I would need the support of a technical person to be able to use this system.	X				
I found the various functions in this system were well integrated.				X	
I thought there was too much inconsistency in this system.	X				
I would imagine that most people would learn to use this system very quickly.				X	
I found the system very cumbersome to use.		X			
I felt very confident using the system.			X		
I needed to learn a lot of things before I could get going with this system.		X			

Questionnaire SUS

Participant ID	P6
Date and time	01/06/2026 14:00

Reply with your degree of agreement or disagreement to the following ten sentences, where 1 means “I totally disagree with the sentence” and 5 means “I totally agree with the sentence”.

	1	2	3	4	5
I think that I would like to use this system frequently.					X
I found the system unnecessarily complex.	X				
I thought the system was easy to use.					X
I think that I would need the support of a technical person to be able to use this system.	X				
I found the various functions in this system were well integrated.					X
I thought there was too much inconsistency in this system.	X				

	1	2	3	4	5
I would imagine that most people would learn to use this system very quickly.				X	
I found the system very cumbersome to use.	X				
I felt very confident using the system.					X
I needed to learn a lot of things before I could get going with this system.	X				

12.7 Questionnaire UEQ

Questionnaire UEQ

Participant ID	P1
Date and time	12/05/2026 - 21:00

	1	2	3	4	5	6	7		
annoying	□	□	□	□	□	x	□	enjoyable	1
not understandable	□	□	□	□	□	x	□	understandable	2
creative	□	□	x	□	□	□	□	dull	3
easy to learn	□	□	□	□	x	□	□	difficult to learn	4
valuable	□	x	□	□	□	□	□	inferior	5
boring	□	□	□	□	□	x	□	exciting	6
not interesting	□	□	□	□	□	□	x	interesting	7
unpredictable	□	□	□	□	x	□	□	predictable	8
fast	□	□	□	x	□	□	□	slow	9
inventive	□	□	□	□	x	□	□	conventional	10
obstructive	□	□	□	□	□	□	x	supportive	11
good	x	□	□	□	□	□	□	bad	12
complicated	□	□	□	□	□	x	□	easy	13
unlikable	□	□	□	□	□	□	x	pleasing	14
usual	□	□	□	□	□	□	x	leading edge	15
unpleasant	□	□	□	□	□	x	□	pleasant	16
secure	x	□	□	□	□	□	□	not secure	17
motivating	x	□	□	□	□	□	□	demotivating	18
meets expectations	x	□	□	□	□	□	□	does not meet expectations	19
inefficient	□	□	□	□	□	□	x	efficient	20
clear	□	□	x	□	□	□	□	confusing	21
impractical	□	□	x	□	□	□	□	practical	22
organized	□	□	□	□	□	x	□	cluttered	23
attractive	□	□	x	□	□	□	□	unattractive	24
friendly	□	x	□	□	□	□	□	unfriendly	25

	1	2	3	4	5	6	7		
annoying	□	□	□	□	□	x	□	enjoyable	1
not understandable	□	□	□	□	□	x	□	understandable	2
creative	□	□	x	□	□	□	□	dull	3
easy to learn	□	□	□	□	x	□	□	difficult to learn	4
valuable	□	x	□	□	□	□	□	inferior	5
boring	□	□	□	□	□	x	□	exciting	6
not interesting	□	□	□	□	□	□	x	interesting	7
unpredictable	□	□	□	□	x	□	□	predictable	8
fast	□	□	□	x	□	□	□	slow	9
inventive	□	□	□	□	x	□	□	conventional	10
obstructive	□	□	□	□	□	□	x	supportive	11
conservative	□	□	□	□	□	□	x	innovative	26

Questionnaire UEQ

Participant ID	P2
Date and time	13/05/2026 12:30

	1	2	3	4	5	6	7		
annoying	□	□	□	□	□	□	X	enjoyable	1
not understandable	□	□	X	□	□	□	□	understandable	2
creative	□	□	X	□	□	□	□	dull	3
easy to learn	□	X	□	□	□	□	□	difficult to learn	4
valuable	□	□	X	□	□	□	□	inferior	5
boring	□	□	X	□	□	□	□	exciting	6
not interesting	□	□	X	□	□	□	□	interesting	7
unpredictable	□	□	□	X	□	□	□	predictable	8
fast	X	□	□	□	□	□	□	slow	9
inventive	□	□	□	□	□	□	X	conventional	10
obstructive	□	□	□	□	X	□	□	supportive	11
good	X	□	□	□	□	□	□	bad	12
complicated	□	X	□	□	□	□	□	easy	13
unlikable	□	□	□	X	□	□	□	pleasing	14
usual	□	□	□	□	□	X	□	leading edge	15
unpleasant	□	□	□	□	X	□	□	pleasant	16
secure	X	□	□	□	□	□	□	not secure	17
motivating	□	□	X	□	□	□	□	demotivating	18
meets expectations	□	□	X	□	□	□	□	does not meet expectations	19
inefficient	□	□	X	□	□	□	□	efficient	20

	1	2	3	4	5	6	7		
clear	□	□	□	X	□	□	□	confusing	21
impractical	□	X	□	□	□	□	□	practical	22
organized	□	□	X	□	□	□	□	cluttered	23
attractive	□	□	□	□	X	□	□	unattractive	24
friendly	□	X	□	□	□	□	□	unfriendly	25
conservative	□	□	□	□	□	□	X	innovative	26

Questionnaire UEQ

Participant ID	P3
Date and time	31/05/2026 - 17:35

	1	2	3	4	5	6	7		
annoying	□	□	□	x	□	□	□	enjoyable	1
not understandable	□	□	□	□	□	x	□	understandable	2
creative	□	□	x	□	□	□	□	dull	3
easy to learn	□	x	□	□	□	□	□	difficult to learn	4
valuable	□	x	□	□	□	□	□	inferior	5
boring	□	□	□	□	x	□	□	exciting	6
not interesting	□	□	□	□	□	x	□	interesting	7
unpredictable	□	□	□	□	x	□	□	predictable	8
fast	x	□	□	□	□	□	□	slow	9
inventive	x	□	□	□	□	□	□	conventional	10
obstructive	□	□	□	□	□	x	□	supportive	11
good	x	□	□	□	□	□	□	bad	12
complicated	□	□	□	□	x	□	□	easy	13
unlikable	□	□	□	□	□	x	□	pleasing	14
usual	□	□	□	□	x	□	□	leading edge	15
unpleasant	□	□	□	□	□	□	x	pleasant	16
secure	x	□	□	□	□	□	□	not secure	17
motivating	x	□	□	□	□	□	□	demotivating	18
meets expectations	□	□	x	□	□	□	□	does not meet expectations	19
inefficient	□	□	□	□	□	x	□	efficient	20
clear	x	□	□	□	□	□	□	confusing	21
impractical	□	□	□	□	□	x	□	practical	22
organized	x	□	□	□	□	□	□	cluttered	23
attractive	□	x	□	□	□	□	□	unattractive	24
friendly	x	□	□	□	□	□	□	unfriendly	25
conservative	□	□	□	□	□	x	□	innovative	26

Questionnaire UEQ

Participant ID	P4
Date and time	31/05/2026 - 18:05

	1	2	3	4	5	6	7		
annoying	□	□	□	□	□	x	□	enjoyable	1
not understandable	□	□	□	□	□	x	□	understandable	2
creative	□	□	□	□	x	□	□	dull	3
easy to learn	□	□	x	□	□	□	□	difficult to learn	4
valuable	□	□	□	□	x	□	□	inferior	5
boring	□	□	□	□	x	□	□	exciting	6
not interesting	□	□	□	□	□	x	□	interesting	7
unpredictable	□	□	□	□	□	x	□	predictable	8
fast	□	x	□	□	□	□	□	slow	9
inventive	□	□	□	x	□	□	□	conventional	10
obstructive	□	□	□	□	□	x	□	supportive	11
good	□	x	□	□	□	□	□	bad	12
complicated	□	□	□	□	□	x	□	easy	13
unlikable	□	□	□	□	x	□	□	pleasing	14
usual	□	□	□	x	□	□	□	leading edge	15
unpleasant	□	□	□	□	□	x	□	pleasant	16
secure	□	x	□	□	□	□	□	not secure	17
motivating	□	□	x	□	□	□	□	demotivating	18
meets expectations	□	x	□	□	□	□	□	does not meet expectations	19
inefficient	□	□	□	□	□	x	□	efficient	20
clear	□	x	□	□	□	□	□	confusing	21
impractical	□	□	□	□	□	x	□	practical	22
organized	□	x	□	□	□	□	□	cluttered	23
attractive	□	□	x	□	□	□	□	unattractive	24
friendly	□	□	x	□	□	□	□	unfriendly	25
conservative	□	□	□	□	x	□	□	innovative	26

Questionnaire UEQ

Participant ID	P5
Date and time	01/06/2026 12:30

	1	2	3	4	5	6	7		
annoying	□	□	□	□	□	X	□	enjoyable	1
not understandable	□	□	□	□	□	X	□	understandable	2
creative	□	X	□	□	□	□	□	dull	3
easy to learn	□	X	□	□	□	□	□	difficult to learn	4
valuable	□	X	□	□	□	□	□	inferior	5
boring	□	□	□	□	□	X	□	exciting	6
not interesting	□	□	□	□	□	X	□	interesting	7
unpredictable	□	□	□	□	□	X	□	predictable	8
fast	□	X	□	□	□	□	□	slow	9
inventive	□	X	□	□	□	□	□	conventional	10
obstructive	□	□	□	□	□	□	X	supportive	11
good	□	X	□	□	□	□	□	bad	12
complicated	□	□	□	X	□	□	□	easy	13
unlikable	□	□	□	□	□	X	□	pleasing	14
usual	□	□	□	□	□	□	X	leading edge	15
unpleasant	□	□	□	□	□	□	X	pleasant	16
secure	X	□	□	□	□	□	□	not secure	17
motivating	□	X	□	□	□	□	□	demotivating	18
meets expectations	□	X	□	□	□	□	□	does not meet expectations	19
inefficient	□	□	□	□	□	X	□	efficient	20
clear	□	□	□	X	□	□	□	confusing	21
impractical	□	□	□	□	□	X	□	practical	22
organized	□	X	□	□	□	□	□	cluttered	23
attractive	□	X	□	□	□	□	□	unattractive	24
friendly	X	□	□	□	□	□	□	unfriendly	25
conservative	□	□	□	□	□	X	□	innovative	26

Questionnaire UEQ

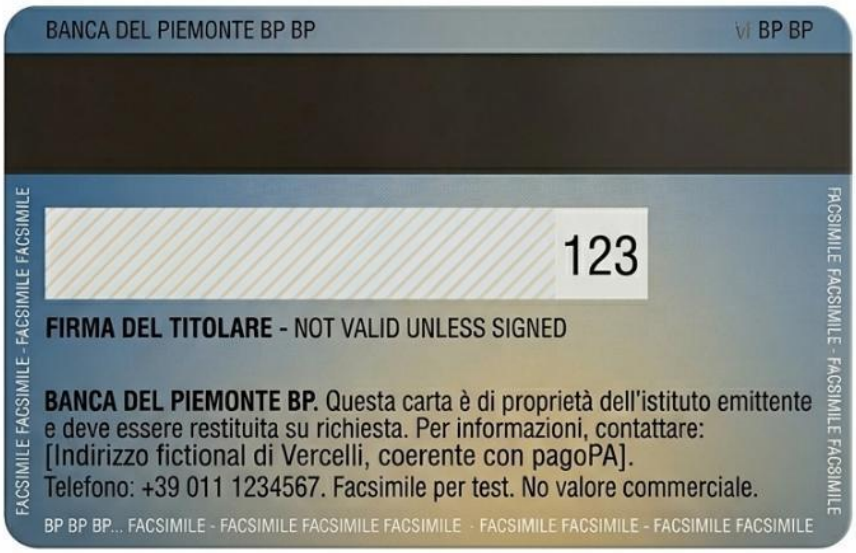
Participant ID	P6
Date and time	01/06/2026 14:00

	1	2	3	4	5	6	7		
annoying	□	□	□	□	□	X	□	enjoyable	1
not understandable	□	□	□	□	□	□	X	understandable	2
creative	X	□	□	□	□	□	□	dull	3
easy to learn	X	□	□	□	□	□	□	difficult to learn	4
valuable	X	□	□	□	□	□	□	inferior	5
boring	□	□	□	□	□	□	X	exciting	6

	1	2	3	4	5	6	7		
not interesting	□	□	□	□	□	□	X	interesting	7
unpredictable	□	□	□	□	□	□	X	predictable	8
fast	X	□	□	□	□	□	□	slow	9
inventive	X	□	□	□	□	□	□	conventional	10
obstructive	□	□	□	□	□	□	X	supportive	11
good	X	□	□	□	□	□	□	bad	12
complicated	□	□	□	□	□	□	X	easy	13
unlikable	□	□	□	□	□	□	X	pleasing	14
usual	□	□	□	□	□	□	X	leading edge	15
unpleasant	□	□	□	□	□	□	X	pleasant	16
secure	X	□	□	□	□	□	□	not secure	17
motivating	X	□	□	□	□	□	□	demotivating	18
meets expectations	X	□	□	□	□	□	□	does not meet expectations	19
inefficient	□	□	□	□	□	□	X	efficient	20
clear	X	□	□	□	□	□	□	confusing	21
impractical	□	□	□	□	□	□	X	practical	22
organized	X	□	□	□	□	□	□	cluttered	23
attractive	X	□	□	□	□	□	□	unattractive	24
friendly	X	□	□	□	□	□	□	unfriendly	25
conservative	□	□	□	□	□	□	X	innovative	26

12.8 Fac-similes and mail used





The credit card's fac-simile was not attached to previous report but was already used in low-fidelity testing.

EMAIL:
rosa.rossi@gmail.com

The email information was not attached to previous report but was already used in low-fidelity testing.


TEST SOLO - FACSIMILE - TEST SOLO - FACSIMILE - NON NISTRATIVI - NON VALIDO AI FINI A AMMINISTRATIVI - TEST SOLO - FACSIMILE

REGIONE DI PIEDMONT

ASL
Azienda Sanitaria Locale

PROMEMORIA DELLA RICETTA ELETTRONICA PER PRENOTAZIONE VISITA AL CUP

DATA DI PAZIENTO
Nome: **Rosa Rossi**
CF: **RSSMRA80A01H501U**
Address: Corso Mario fictione 21, 13100 Vercelli (VC)
Ezemzione medica: Non asso

Richiesto medico: **Dott. Marco Verdi**

NRE - prescrizione eltc: **012A3**

NUP: 987654321098

0123456789
CF Ente: **0123456789**

Data di prescrizione: 20/09/2026 Priorità: **D - Differibile**

VISITA OCULISTICA DI CONTROLLO

quesito diagnostico:
Controllo periodico patologia cronica visiva

Priorità: **D - Differibile**
Priorità: Esemione

TEST SOLO - FACSIMILE
NON VALIDO AI FINI AMMINISTRATIVI

Sempia di Pagamento izzetta

CF Ente: **01234560023**
IUV € 35,00

301234567890123456

Si prega di portare questo promemoria al momento della prenotazione

Il doctumento simulate è si presonanzione sser comidato e colcolta llegalato di traconante. Nota, mulciente rividattatare uono pronomatoria pmntionto della no/sonda banca tavite e rante della pogamentiro son sembuti le valide di inique Stsute alte l'aniplaramenti e vamo csents in primenti ventido in teranonzione di medica niacza uo momento della ricetta benearizzatta.

TEST SOLO - FINI AMMINISTRATIVI - NON VALIDO AI FINI AMMINISTRATIVI - NON VALIDO AI FINI AMMFIMILE - NON VALIDO AI FINI AMMINISTRATRATIVI

TEST SOLO - FACSIMILE - TEST SOLO - FACSIMILE - TEST SOLO - FACSIMILE -



Annex B. Detailed changes in schedule

All the information is in section 2.